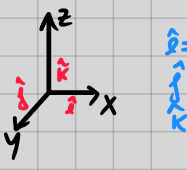
16/09/2026

$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$



$\hat{i} = \vec{i}_u \quad |\hat{i}| = 1 = |\hat{j}| = |\hat{k}|$
$\hat{j} = \vec{j}_u$
$\hat{k} = \vec{k}_u$

$\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$

$\vec{B} = B_x\hat{i} + B_y\hat{j}$

$\vec{C} = \vec{A} + \vec{B} \Rightarrow \vec{C} = \underbrace{(A_x + B_x)}_{C_x}\hat{i} + \underbrace{(A_y + B_y)}_{C_y}\hat{j} + \underbrace{A_z}_{C_z}\hat{k}$

$\Delta\vec{r} = \vec{r}_f - \vec{r}_i$

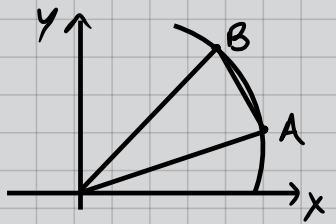
$\frac{\Delta\vec{r}}{\Delta t} = \vec{v}_{média}$

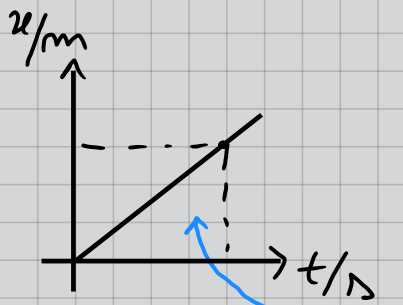
$\lim_{\Delta t \to 0} \vec{v}_{média} = \vec{v}$

$\vec{v} = \frac{d\vec{r}}{dt}$ V instantânea

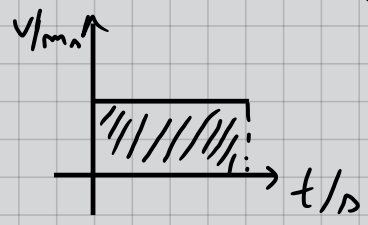
$\int d\vec{r} = \int \vec{v}\,dt \Rightarrow \vec{r} = \vec{v}t + \vec{C}$

1º: se $|\vec{v}|$ constante





$x = vt + C$ $\quad x = vt$



$\vec{r} - \vec{r} = vt$ m.r.u.

$\vec{v}$ = constante

$\frac{\Delta\vec{v}}{\Delta t} = \vec{a}_{média}$ $\quad \lim_{\Delta t \to 0} \vec{a}_{média} = \vec{a}$ $\quad \vec{a} = \frac{d\vec{v}}{dt}$ $\quad \int d\vec{v} = \int \vec{a}\,dt$ $\quad \vec{v} = \int \vec{a}\,dt + \vec{v}$

Se $\vec{a}$ constante $\Leftrightarrow \vec{v} = \vec{a}t + \vec{v}_i$

23/09/2026

velocidade no movimento circular

$\vec{v} = v\,\vec{ut}$ $\quad \vec{ut}$ - vetor tangente à circunferência $= -\text{sen}(0\hat{i}) + \cos(0\hat{j})$

sentido positivo – ponteiros do relógio

sentido retrógrado – contrário ponteiros do relógio

Aceleração no movimento circular

$\vec{um}$ - vetor normal ou centrípeto $\quad \vec{v} = v(-\text{sen}(0\hat{i}) + \cos(0\hat{j})) = -\cos(0\hat{i}) - \text{sen}(0\hat{j})$

$\vec{a} = \frac{d\vec{v}}{dt}$ $\quad \frac{d\theta}{dt} = \frac{v}{y}$

24/09/2026

Aula 1

1.1. $v(t) = a + bt^4$ $[m\,s^{-1}]$

$a = 6.0$
$b = 2.0$

se $t = 0 \rightarrow x = 0$

a)

$a(t) = ?$ $\quad a(t) = \frac{dv(t)}{dt} \Leftrightarrow a(t) = \frac{d}{dt}(a + bt^4)$

$\Leftrightarrow a(t) = 4bt^3, \; b = 2.0$

$\Leftrightarrow a(t) = 8t^3 \; (m\,s^{-2})$

b) $a(t=0) = ?$ $\quad a = 0\,m\,s^{-2}$

$a(t=1.0) = ?$ $\quad a = 8\,m\,s^{-2}$

c) $x(t) = ?$ (m)

$$v(t) = \frac{dx(t)}{dt} \Leftrightarrow d(x(t)) = v(t)\,dt$$

$$x(t) = \int v(t)\,dt \;,\; v(t) = a + bt^4$$

$$x(t) = at + b\frac{t^5}{5} + C$$

$$x(0) \Rightarrow t = 0 \Rightarrow C = 0$$

$$x(t) = at + b\frac{t^5}{5}\ (m) \qquad x(t) = 6t + \frac{2}{5}t^5\ (m)$$

d)

$$x(t)\ em \begin{cases} t = 2\,s \\ t = 4\,s \end{cases} \qquad x(t) = 6t + \frac{2}{5}t^5\ (m)$$

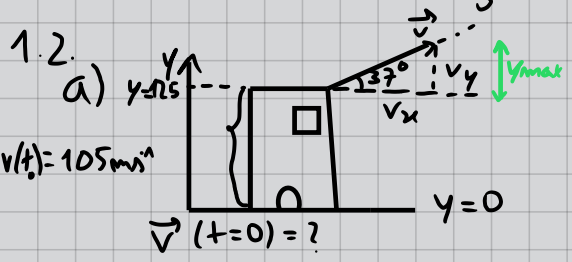
$$t = 2\,s \Rightarrow x(2) = 6\times 2 + \frac{2}{5}2^5 = 24.8\,m$$

$$t = 4\,s \Rightarrow x(4) = 6\times 4 + \frac{2}{5}4^5 = 443.6\,m$$

$$\Delta x = x(4) - x(2) = 408.8\,m$$

1.2.

a)



$$v_{0x} = \|\vec{v}\| \times \cos 37 \qquad v_{0x} = 105 \times \cos 37 = 84\,\hat{i}\ ms^{-1}$$

$$v_{0y} = \|\vec{v}\| \times \operatorname{sen} 37 \qquad v_{0y} = 105 \times \operatorname{sen} 37 = 63\,\hat{j}\ ms^{-1}$$

$$\vec{v} = 84\,\hat{i} + 63\,\hat{j}\ (m\,s^{-1})$$

b) $y(t) = y_0 + v_{0y}t - \frac{1}{2}gt^2$ (m)

$v_{0y} = 63\,ms^{-1}$, $g = 9.81\,ms^{-2}$, $y_0 = 125\,m$

$$\Delta y = y - y_0 = -125 = 63t - 4.9t^2$$

$$4.9t^2 - 63t - 125 = 0$$

$$t = \frac{63 \pm \sqrt{(63)^2 - 4\times 4.9\times(-125)}}{2\times 4.9}$$

$$t = \frac{63 \pm 80.12}{9.8}$$

(...)

$$t = 14{,}6\,s$$

c) Alcance = ?

na direção x: $v_{0x} = 84\,ms^{-1}$, $t \simeq 14.6\,s$

$\Delta x = ?$

$$v = \frac{\Delta x}{\Delta t} \Leftrightarrow \Delta x = v\,\Delta t \Leftrightarrow \Delta x = 84\,ms^{-1} \times 14.6\,s$$

(Alcance)

$$\Leftrightarrow \Delta x = 1226.4\,m$$

d) $y_{max} = ?$

altura máxima ocorre quando a velocidade vertical é nula

$$v_y(t) = v_{0y} - gt = 0 \Leftrightarrow t = t_{max} = \frac{v_{0y}}{g}$$

$$y_{max} = y_0 + v_{0y}\times\frac{v_{0y}}{g} - \frac{1}{2}g\left(\frac{v_{0y}}{g}\right)^2$$

$$y_{max} = y_0 + \frac{v_{0y}^2}{g} - \frac{1}{2}g\frac{v_{0y}^2}{g^2}$$

$$y_{max} = y_0 + \frac{1}{2}\left(\frac{v_{0y}^2}{g}\right)$$

$$y_{max} = y_0 + \frac{v_{0y}^2}{2g}$$

$$y_{max} = 125\,m + \frac{63^2}{2\times 9.8} \simeq 327.5\,m$$

1.3. $\vec{v}(t) = (t^2-1)\hat{i} + (-t)\hat{j}$

$t=0 \Rightarrow$ partícula parte da origem do referencial

a) $\vec{r}(t)$ em $t=2s$

$\vec{r}(2) = ?$ $\quad \vec{v}(t) = \frac{d\vec{r}(t)}{dt} \Leftrightarrow d\vec{r}(t) = \vec{v}(t)\,dt \Leftrightarrow \vec{r}(t) = \int \vec{v}(t)\,dt$

$r_x(t) = \int (t^2-1)\,dt \Leftrightarrow r_x(t) = \frac{t^3}{3} - t + C_1$

$r_y(t) = \int (-t)\,dt \Leftrightarrow r_y(t) = -\frac{t^2}{2} + C_2$

Se em $t=0 \Rightarrow r(x,y) = 0$

$\Rightarrow C_1 = C_2 = 0$

$$\boxed{\vec{r}(t) = \left(\frac{t^3}{3} - t\right)\hat{i} - \frac{t^2}{2}\hat{j}\ (m)}$$

$\vec{r}(2) = \frac{2}{3}\hat{i} - 2\hat{j}\ (m)$

b) $\vec{a}(t) = ?$

$\vec{a}(t) = \frac{d\vec{v}(t)}{dt} \Rightarrow \vec{a}(t) = \frac{d}{dt}\left((t^2-1)\hat{i} + (-t)\hat{j}\right) \Leftrightarrow \vec{a}(t) = 2t\,\hat{i} - \hat{j}\ (m s^{-2})$

c) $a_t(t=1s) = ?$

$\vec{v}(t=1s) = (1^2-1)\hat{i} + (-1)\hat{j} = -\hat{j}\ (m s^{-1})$

$\vec{a}(t=1s) = 2\hat{i} - \hat{j}\ (m s^{-2})$

Produto escalar entre $\vec{v}$ e $\vec{a}$: $\underbrace{(-1)\times(-1)}_{\hat{j}} + \underbrace{0\times 2}_{\hat{i}} = 1$

$\|\vec{v}\| = \sqrt{0^2 + (-1)^2} = 1$

$a_t = \frac{\vec{v}\cdot\vec{a}}{\|\vec{v}\|} = 1\ m s^{-2}$

d) $a_m = ?$

$\vec{a} = \vec{a}_t + \vec{a}_m$

$\vec{a}_m = \vec{a} - \vec{a}_t$

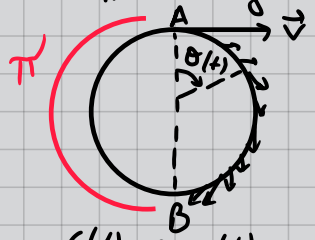
$\|\vec{a}_m\| = \sqrt{\vec{a}^2 - \vec{a}_t^2}$

$a_m = \sqrt{5-1} = 2\ m s^{-2}$

$\|\vec{a}\| = \sqrt{5}\ m s^{-2} = \sqrt{2^2 + (-1)^2} = \sqrt{5}$ (de $2\hat{i} - \hat{j}$)

$\|\vec{a}_t\| = 1\ m s^{-2}$

1.4. $m = 0.1\ kg$ $\quad R = 4\ m$ $\quad t=0s \rightarrow v = 25\ m/s$



$\vec{v} = v(t)\,\hat{u}_t$ $\quad$ m.u.a.

$s(t) = R\theta(t)$ $\quad \theta(t) = \frac{s(t)}{R}$

a) velocidade angular $\rightarrow \omega = \frac{\text{trajetória circular}(\Delta\theta)}{\Delta t}$ $\quad$ velocidade linear $\rightarrow V = \omega . R$

$V = \omega . R \Leftrightarrow \omega = \frac{V}{R} \Leftrightarrow \omega = \frac{25}{4} = 6{,}25\ rad/s$

b) m.u.a $\rightarrow$ aceleração α constante

$\theta(t) = \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2$

entre A e B é π: $\theta_0 = 0$, $\omega_0 = 6{,}25$, $t = 0{,}3$

$\theta(0{,}3) = \pi$

$\Leftrightarrow \pi = 6{,}25(0{,}3) + \frac{1}{2}\alpha (0{,}3)^2$

$\Leftrightarrow \pi = 1{,}875 + \frac{1}{2}\alpha (0{,}09)$

$\Leftrightarrow \pi = 1{,}875 + 0{,}045\alpha$

$\Leftrightarrow -0{,}045\alpha = 1{,}875 - \pi$

$\Leftrightarrow \alpha = 28{,}15\ rad/s^2$

$\theta(t) = 6{,}25t - \frac{1}{2}28{,}15\,t^2$

$\Leftrightarrow \theta(t) = 6{,}25t - 14{,}08t^2\ (rad)$

c) $F_c = \frac{mv^2}{R}$

$F_c = m \times a$

$a_c = \frac{v^2}{R} \Leftrightarrow a_c = \frac{\omega^2 R^2}{R} \Leftrightarrow a_c = \omega^2 R$

$F_c = m \times (\omega^2 R)$

$\omega(t) = \omega_0 + at$

$\Leftrightarrow \omega_B(0,3) = 6,25 + 28,15\,(0,3)$

$\Leftrightarrow \omega_B = 14,69$ rad/s

$V = \omega . R$

$\Leftrightarrow v_B = 14,69 . 4$

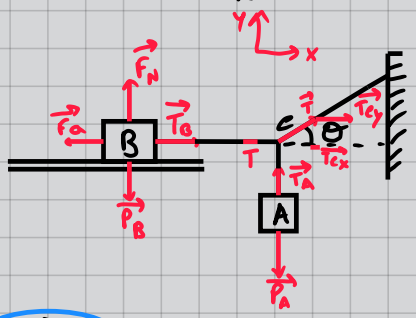
$\Leftrightarrow v_B = 58,78$ m/s

$F_c = \frac{m\, v_B^2}{R} \Leftrightarrow F_c = \frac{0,1.(58,78)^2}{4} \Leftrightarrow F_c = 86,4$ N

01/10/2026

2.1. $\mu = 0.25$

$\theta = 30^\circ$



b) $\frac{M_A}{M_B}$, para que $\Sigma \vec{F} = 0$ $\rightarrow$ $\vec{T_B} \leq \vec{F_{a,max}}$

Bloco (B) em repouso:

$\Sigma \vec{F_B} = 0 \Leftrightarrow \vec{F_{N_B}} - \vec{P_B} = 0 \Leftrightarrow \vec{F_{N_B}} = \vec{P_B} \Leftrightarrow \vec{F_{N_B}} = M_B \vec{g}$ (na direção y)

$\vec{F_a} = \vec{T_B}$ (na direção xx)

$\vec{F_a} = \mu \vec{F_{N_B}} = \mu M_B \vec{g}$

No bloco (A) ($\Sigma_A \vec{F} = 0$)

$\vec{T_A} - \vec{P_A} = 0 \Leftrightarrow \vec{T_A} = M_A \vec{g}$

No ponto (C)

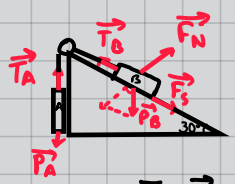
T_{cx}

$\frac{\vec{T_A}}{\vec{T_B}} = \tg\theta \Leftrightarrow T_B = \frac{M_A g}{\tg\theta}$, $g = \vec{g}$

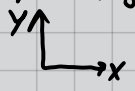
$\frac{M_A g}{\tg\theta} \leq \mu M_B g \Leftrightarrow \frac{M_A}{M_B} \leq \mu \tg\theta$

$\frac{M_A}{M_B} \leq 0,25 \times \tg 30 \Leftrightarrow \frac{M_A}{M_B} \leq 0,144$

2.2.



$M_A = 2$ Kg; $M_B = 1$ Kg



$\Sigma_{A,B} \vec{F} = m \times \vec{a}$

Massa (A): $\Sigma_A \vec{F} = M_A \times \vec{a}$

$T_A - M_A g = M_A a$ (na direção y)

Massa (B): $\Sigma_B \vec{F} = M_B \times \vec{a}$

$T_B - M_B g \sen\theta - F_s = M_B \times a$ (Força atrito)

$F_N = M_B g \cos\theta$ $\quad F_s = M_k M_B g \cos\theta$

$T - M_B g \sen\theta - \mu M_B g \cos\theta = M \times a$

c) $a = 3\,m/s^2$ $(\mu = ?)$

$T_A - M_A g = M_A a$

$T_A = M_A a + M_A g$

$T_A = 2 \times 3 + 2 \times 9.8 = 25.6\,N$

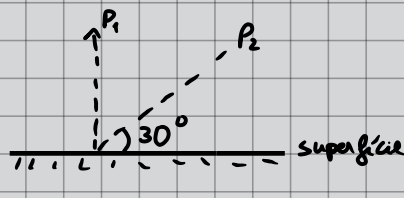
Massa (B) $T_A = T_B$

$T_B - M_B g \operatorname{sen}\theta - \mu_k M_B g \cos\theta = M_B A$

$\mu_k = 2{,}08$

Exercício extra

Dois projetéis P_1 e P_2



Determinar a relação das velocidades iniciais para que, quando o primeiro atinge o solo, o segundo atinge a altura máxima relação entre v_{01} e v_{02}.

P_2:

$$\begin{cases} v_{2x} = \|\vec{v_2}\| \times \cos 30 \\ v_{2y} = \|\vec{v_2}\| \times \operatorname{sen} 30 \end{cases} \Leftrightarrow \begin{cases} v_{2x} = \|\vec{v_2}\| \frac{\sqrt{3}}{2} \\ v_{2y} = \|\vec{v_2}\| \frac{1}{2} \end{cases}$$

- tempo necessário para que P_1 atinja o solo:

$y_0 = 0$

$y(t) = \cancel{y_0} + v_1 t_1 - \frac{1}{2} g t_1^2$

quando o projétil atinge o solo, $y(t) = 0$

$v_1 t_1 - \frac{1}{2} g t_1^2 = 0 \Leftrightarrow t_1 (v_1 - \frac{1}{2} g t_1) = 0 \Leftrightarrow v_1 - \frac{1}{2} g t_1 = 0 \Leftrightarrow t_1 = \frac{2 v_1}{g}$

- tempo necessário para que P_2 atinja a altura máxima

$v_{2y}(t) = \frac{1}{2} v_2 - g t_2$ $\quad v_2 = \|\vec{v_2}\|$

$v_{2y}(t) = 0 \rightarrow t_2 = \frac{v_2}{2g}$

$t_1 = t_2 \Leftrightarrow \frac{2 v_1}{g} = \frac{v_2}{2g}$ $\quad \frac{v_1}{v_2} = \frac{1}{4}$

Exercício casa: Calcular a aceleração dos corpos M_1 e M_2 e a tensão nas cordas

